

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2023

ESE 4115

(Power Plant Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | Distinguish between ideal load curve and realized load curve. | 06 |
| 1(b). | What is meant by base load power plant and peak load power plant? What types of power plant are suitable for peak load power plant? | 09 |
| 1(c). | What factors should be considered while selecting the capacity of a power plant? | 05 |
| 1(d). | What are the advantages of interconnected grid system? | 06 |
| 1(e). | A generation station has a maximum demand of 25 MW, a load factor of 60%, a plant capacity factor of 50%, and a plant use factor of 74%. Find (i) the reserve capacity of the plant, (ii) the daily energy produced, and (iii) maximum energy that could be produced daily if the plant while running as per schedule, were fully loaded. | 09 |
| | | |
| 2(a). | What are the effects of variable load power plant design and operation? | 08 |
| 2(b). | What is meant by depreciation of power plant? What factors influence the depreciation? | 08 |
| 2(c). | What is meant by meter rate? What are the different types of meter rate in commonly use? | 08 |
| 2(d). | Calculate the generating cost per kWh delivered from a power station using the following data:
Plant capacity = 100 MW
Annual load factor = 45%
Capital cost = Tk 105 crores
Annual cost of wages, taxation etc. = Tk 40 lac
Cost of fuel, lubrication, maintenance etc. = Tk 1/kWh generated
Interest = 4% per annum
Depreciation = 6% per annum of initial value | 11 |
| | | |
| 3(a). | What are the purposes of load forecasting? | 07 |
| 3(b). | What are the methods of short term load forecasting? Explain the similar-day approach of load forecasting. | 08 |
| 3(c). | Explain the present situation of power generation in Bangladesh from renewable energy sources. | 09 |
| 3(d). | According to the power system master plan 2015, describe the future demand of electricity in Bangladesh. How can the demand be met? Explain. | 11 |
| | | |
| 4(a). | Define following terms related to hydroelectric power plant:
(i) Surge tank, (ii) Spillway, (iii) Hydrograph. | 09 |
| 4(b). | Classify hydroelectric power plant. | 08 |

- 4(c). Which factors should be considered while selecting the turbines for hydro-electric power plant? Explain. 07
- 4(d). The following data refers to a proposed hydroelectric power plant: 11
- Catchment area = 800 km²
 Rainfall = 185 cm/year
 Available head = 35 m
 Infiltration and evaporation loss = 42%
 Turbine efficiency = 84%
 Generator efficiency = 85%
 Load factor = 0.5.
- (i) Calculate power developed, and (ii) Suggest suitable turbines for this plant.

SECTION – B

- 5(a). With neat sketch, describe the working principle of a diesel power plant. 15
- 5(b). What are the applications of diesel power plant? Explain. 10
- 5(c). What are the methods used to start a diesel power plant? Briefly explain. 05
- 5(d). Write down the necessity of cooling system in diesel power plant. 05
- 6(a). What are the advantages of gas turbine power plant as a peak load plant? 07
- 6(b). Draw a layout of an open cycle gas turbine power plant. 08
- 6(c). What is meant by combined cycle power plant? What are the advantages of these types of power plant? 10
- 6(d). Show that the net work output of a gas turbine power plant will be maximum when the exit temperature of working fluid from compressor and turbine are same. 10
- 7(a). Explain the effect of regeneration on the work output and efficiency of a gas turbine power plant. 12
- 7(b). With neat sketch, describe the working principle of steam power plant. 15
- 7(c). What are the advantages of steam power plant as a base load power plant? 08
- 8(a). Why treatment of raw water is necessary before using it as feed water in boiler? Explain the working principle of Zeolite process for softening the feed water. 11
- 8(b). What is meant by draught? Write down its importance in power plant. 07
- 8(c). Write short notes on the following terms related to cooling tower: 09
- (i) approach, (ii) range, and (iii) cooling efficiency.
- 8(d). A chimney is 28 m height and the temperature of hot gases inside the chimney is 320 °C. The temperature of outside air is 27 °C and furnace is supplied with 15 kg of air per kg of coal burnt. Calculate (i) draught in mm of water, and (ii) draught height in meters of hot gases. 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2023

ESE 4025

(HVAC&R System Design)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks. Required graph /table will be provided

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | Describe the working principle of vapor compression refrigeration system with neat sketch. | 10 |
| 1(b). | Explain the effects of evaporating and condensing temperature on the performance of vapor compression refrigeration cycle. | 10 |
| 1(c). | How the performance of a practical refrigeration is evaluated? Describe. | 08 |
| 1(d). | Calculate the refrigeration tonnage for 1500 kg ice from 0 °C to 25 °C. | 07 |
| | | |
| 2(a). | Why multi-stage vapor compression is required? Write down the advantages and disadvantages of a multi-stage vapor compression system. | 10 |
| 2(b). | Describe two-stage expansion system with a flash cooler with necessary diagrams. | 13 |
| 2(c). | An Ammonia refrigerator produces 25 tonnes of ice from and at 0 °C in 24 hours. The temperature range of the compressor is from 15 °C to –5 °C. The vapor is dry saturated at the end of compression and an expansion valve is used. Assume a COP to 45% of the theoretical value. Now calculate the power required to drive the compressor. Latent heat of ice is 335 kJ/kg. | 12 |
| | | |
| 3(a). | Write down the classification of vapor absorption refrigeration system. Describe the triple effect vapor absorption system with a schematic diagram. | 13 |
| 3(b). | Discuss the environmental and safety properties to consider before selecting a refrigerant. | 08 |
| 3(c). | What is meant by hermetic sealed compressor? Write down its advantages and disadvantages. | 08 |
| 3(d). | Differentiate between VCR and VAR systems. | 06 |
| | | |
| 4(a). | Describe the working principle of an air refrigerator system. Also mention the necessity of cooling for an aircraft. | 10 |
| 4(b). | Explain the thermostatic expansion valve with a neat sketch. | 10 |
| 4(c). | Write down the differences between air cooled condenser and water cooled condenser. | 05 |
| 4(d). | How thermo-electric cooling technique differs from solar-assisted cooling system? Explain which one is more efficient system and why? Describe. | 10 |

SECTION – B

- | | |
|---|----|
| 5(a). Define the following terms:
(i) Specific humidity,
(ii) Degree of saturation,
(iii) Wet bulb temperature,
(iv) Humid specific heat. | 12 |
| 5(b). What is meant by SHF? How SHF line is drawn in psychrometric chart? | 06 |
| 5(c). Explain the working principle of an air washer. | 06 |
| 5(d). Describe a summer air-conditioning with ventilation air-bypass factor. | 11 |
| | |
| 6(a). Draw the ASHRAE comfort chart. | 05 |
| 6(b). What is meant by indoor design conditions and outdoor design conditions? | 06 |
| 6(c). Describe the various all-air central air-conditioning system. | 12 |
| 6(d). What are effects of IAQ on health and comfort? How to control IAQ? | 12 |
| | |
| 7(a). Write down the functions of the followings: (i) Dampers; (ii) Diffusers | 06 |
| 7(b). Distinguish between multi-split system and VRF system. | 07 |
| 7(c). What are assumptions should be considered while calculating the cooling load. | 07 |
| 7(d). Define heating and cooling load. Describe how the cooling load in the form of grand total load on air conditioning system can be evaluated. | 15 |
| | |
| 8(a). What is meant by sol-air temperature? | 04 |
| 8(b). What factors should be considered while selecting the location of supply air outlets? | 07 |
| 8(c). Write down the classifications of cooling towers? Also describe how the performance of cooling tower is evaluated? | 09 |
| 8(d). Determine the size of the duct system shown in figure using equal friction method assuming pressure drop of 0.1 mm H ₂ O/m length. | 15 |

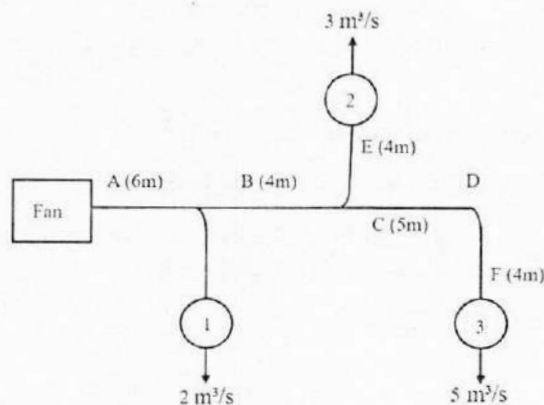


Figure of Question No. 8(d)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2023

ESE 4105

(Fuel Combustion and IC Engines)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|--|----|
| 1(a). | Briefly explain the measurement methods and significance of the following properties:
(i) API gravity, (ii) Reid vapor pressure, (iii) Cetane number, (iv) Octane number | 12 |
| 1(b). | What are the key differences between proximate analysis and ultimate analysis of fuel? | 06 |
| 1(c). | How does the octane number correlate with the performance of IC engines? | 06 |
| 1(d). | How do the storage and handling requirement of solid, liquid and gaseous fuels differ and what safety considerations are associated with each? | 11 |
| 2(a). | Why cooling and lubricating systems are necessary for IC engines? | 05 |
| 2(b). | How does a liquid cooling system differ from an air cooling system in terms of applications in IC engines? | 10 |
| 2(c). | How does the use of a thermostat in a water cooling system regulate engine temperature and what happens if the thermostat fails to function properly? | 10 |
| 2(d). | Describe the water cooling system of IC engine with a schematic diagram. | 10 |
| 3(a). | What is carburation in an IC engine? Describe the working principle of a carburator in IC engine. | 08 |
| 3(b). | How does the air-fuel mixture requirement change between idling, cruising, and full throttle conditions in IC engine? | 10 |
| 3(c). | Describe the different types of fuel injection systems and injector nozzles. | 10 |
| 3(d). | An engine requires 5 kg of air per minute from its carburator. The air is at a pressure of 1.05 bar and a temperature of 25 °C. Determine the throat diameter of the venturi when the air-flow velocity at the throat is 1.02 m/s with a velocity coefficient of 0.8. Assume isentropic flow and consider air as compressible. | 07 |
| 4(a). | Describe the pollutants formation mechanisms in SI engine with appropriate diagrams. | 11 |
| 4(b). | What is particulate emission? How can particulate emission be measured? Why does particulate emission from CI engines differ significantly than that of SI engines? | 12 |
| 4(c). | How do catalytic converter in IC engines reduce the emissions of hydrocarbon (HC), carbon monoxide (CO), and nitrogen oxides (NO _x), and what are the chemical reactions involved? | 12 |

SECTION – B

- 5(a). What is the physical significance of volumetric efficiency? Draw and explain the following engine performance curves: 14
(i) Brake torque, (ii) Brake specific fuel consumption, (iii) volumetric efficiency.
- 5(b). With the help of indicator diagram, explain: 06
(i) Net indicated work per cycle
(ii) Pumping work per cycle.
- 5(c). What is meant by “Equivalence ratio”? “Generally a CI engine operates with lean mixture”- Explain. 09
- 5(d). How does pumping work in a turbocharged engine differ from that in a naturally aspirated engine? Discuss. 06
- 6(a). Discuss the different stages of combustion in a typical SI engine with suitable diagrams. 10
- 6(b). What are meant by surface ignition? “Pre-ignition is more harmful than post-ignition”- Discuss. 07
- 6(c). How do ignition delay and knock in SI engine differ from those in CI engine? Explain with appropriate diagrams. 12
- 6(d). “CI engine performance is smoke limited”-Justify the statement. 06
- 7(a). Discuss the three strategies of mixture formation process in SI engine with suitable diagrams. 12
- 7(b). “A good SI engine fuel is a bad CI engine fuel”- Explain with appropriate diagrams. 10
- 7(c). Discuss the major factors that influence the tendency of knock in CI engine. 13
- 8(a). Describe the effect of supercharging. Which engine is more suitable for supercharging among SI engine or CI engine? Explain why? 11
- 8(b). What types of engine modifications are required if supercharging is to be employed? Explain. 08
- 8(c). Describe the different stages of combustion in a CI engine with the help of rate of heat release (ROHR) verses crank angle (θ) diagram. 11
- 8(d). Why poor ignition quality fuel can be used in low speed large sized CI engine? Explain. 05

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2023

ESE 4007

(Smart Grid Technology)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks. Required graph /table will be provided
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Make a comparison between the conventional power system and a smart grid. 10
- 1(b). What are the differences between an active and a passive distribution network? Draw the block diagram of an active distribution system and describe its operation. 10
- 1(c). Provide an overview of the following technologies required for smart grids: 15
- (i) Information and communication technologies,
 - (ii) Sensing, measurement, control and automation technologies,
 - (iii) Power electronics and energy storage.
- 2(a). Define the term “Distributed Generation (DG)”. What are the advantages of DG? 10
- 2(b). Draw the typical configuration of a micro-grid. Describe the grid-connected and islanded operation of the micro-grid. 10
- 2(c). What is meant by SCADA? What are the advantages of a SCADA integrated system? Show the possible communication infrastructure for the smart grid including HAN, NAN, and WAN. 15
- 3(a). What are the differences between AMR and AMI? How do AMI meters work? 10
- 3(b). What is meant by demand side management? Describe how the following techniques modify the load consumptions patterns: 15
- (i) Load shifting, (ii) Valley filling, and (iii) Peak clipping.
- 3(c). What are the key features of a phasor measurement unit (PMU)? Explain the operation of a PMU with necessary block diagrams. 10
- 4(a). Why is the contingency analysis important for power system studies? Describe the term “security-constrained optimal power flow” with an example. 10
- 4(b). Write down the necessity of GIS and google mapping tools in smart grids. 05
- 4(c). Write down the name of the classical methods of load flow solution. What are the essential features of the load flow solution in smart grids? 10
- 4(d). Design a decentralized meter reading system using the multi-agent approach. 10

SECTION – B

- 5(a). What are the bus and infinite bus in power system? Describe a method to derive bus admittance matrix in smart power system and prove that $I_{bus} = Y_{bus}V_{bus}$, where I_{bus} is the bus current, V_{bus} is the bus voltage and Y_{bus} is bus admittance matrix. 15
- 5(b). How can the maximum power output (P_{max}) of a generator be effectively improved? What are the primary functions of a thyristor-based SVC (Static Var Compensator) in power system? Explain with diagram. 13
- 5(c). Why is power quality management in smart grid so important? 07
- 6(a). In power system transient stability analysis, what principle does the Equal Area Criterion (EAC) assess and describe with necessary schematic diagram. 12
- 6(b). Write a short note on vehicle-to-grid (V2G) technology. 08
- 6(c). Draw and discuss the simple feedback loop block diagram of generator and load. What are the primary methods and devices used to control reactive power in power systems? And How do AVR (Automatic Voltage Regulator) maintain the voltage stability in smart grid? 15
- 7(a). What are some current research challenges in integrating renewable energy sources into the smart grid infrastructure? How does education and training contribute to the successful implementation of smart grid? 12
- 7(b). What is meant by ANN? How does the ANN facilitate the assessment of smart grid performance? 10
- 7(c). What is interoperability in smart grid environment? How can false data injection impact the stability and reliability of smart grid operations, and what measures can be taken to mitigate these risk effectively? 13
- 8(a). What are the primary barriers to the development of smart grids, and what solutions can be implemented to overcome these obstacles? Discuss. 13
- 8(b). How do expert systems (ES) enhance decision-making and operational efficiency in smart grid? Discuss. 10
- 8(c). Mention the names of different voltage stability assessment techniques. Describe one of them. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2023

ESE 4125

(Safety and Environmental Aspects of Energy Projects)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

iv) Table, chart may be provided on request.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | Define and explain the followings: | 12 |
| | (i) Hazard, (ii) Occupational accident, (iii) Occupational safety, (iv) Safety management. | |
| 1(b). | How do occupational safety, health and environmental management practices intersect to ensure comprehensive industrial workplace safety? Explain. | 07 |
| 1(c). | Briefly explain the thermal hazards that workers face in extreme temperature conditions? Also discuss their effects? | 10 |
| 1(d). | What are the psychosocial hazards and how do they affect workplace safety? | 06 |
| 2(a). | Briefly explain the main methods of protection against electrical hazards in industry. | 12 |
| 2(b). | Define fire and flame. Briefly explain the fire-fighting procedure and fire-fighting materials for each class of fire. | 13 |
| 2(c). | Briefly explain the followings: | 10 |
| | (i) Heat cramps, (ii) Cave in, (iii) Asphyxiation, (iv) Frostbite, (v) Heat stroke | |
| 3(a). | Define 'Risk Management'. Briefly explain the key steps in the hazard identification process. | 10 |
| 3(b). | What is a bow-tie diagram and how is it used in risk management? Explain. | 10 |
| 3(c). | What is the hierarchy of mitigation controls for managing workplace hazards and how does different types of PPE fit into it? | 10 |
| 3(d). | Differentiate between Preliminary Hazard Analysis (PHA) and HAZOP Analysis. | 05 |
| 4(a). | How does Heinrich's Domino theory explain the causation of accidents in the workplace? | 07 |
| 4(b). | What are the purposes of accident report analysis? Also explain the terms: AFR, ASR and FAR with their significance. | 07 |
| 4(c). | What is the Triple Bottom Line (TBL) framework? Who are the typical members of a safety committee and what are the roles do they play? | 07 |
| 4(d). | What are the steps for emergency preparedness plan? Briefly explain them. | 14 |

SECTION – B

5(a).	Define occupational health and occupational diseases. What are the function of occupational health service?	12
5(b).	Explain the followings: (i) Urticaria, (ii) Pigmentary changes, (iii) Blood disorders, (iv) Sore eye.	08
5(c).	Distinguish between inhalation, ingestion and injection.	06
5(d).	Define occupational ergonomics. Mention five examples of good occupational ergonomics and explain them.	09
6(a).	What is meant by hazardous chemicals? Explain the hazardous chemical dose-response relationship. Also point out (show) the followings in this relationship: (i) Lethal, (ii) Sub-lethal, (iii) Acute.	08
6(b).	Why is necessary to study of PEL and TLV? Explain the relation between them.	07
6(c).	What is the primary purpose of Material Safety Data Sheet (MSDS) and why it is important for work safety?	06
6(d).	Discuss about the following terms to prepare a MSDS: (i) Identification of the substance (ii) Hazards identification (iii) First aid measures (iv) Handling and storage (v) Toxicological information (vi) Regulatory information	14
7(a).	What are the major sources of water pollution? How to control them?	08
7(b).	Explain the following environmental impact and mention their main causes: (i) Local, (ii) Regional, (iii) Global.	09
7(c).	What is meant by sustainable development? Briefly discuss about the challenges of sustainable development.	08
7(d).	What are the major greenhouse gases that are responsible for global warming? Briefly explain the mechanism of greenhouse effect to create global warming.	10
8(a).	Write short notes on (i) Acid rain, and (ii) SMOG.	06
8(b).	When IEE and EIA studies are necessary? required for a project? Explain the EIA process in details.	10
8(c).	What is Environmental Management Plan (EMP)? In which stage of a new project, EMP is necessary and why?	06
8(d).	Discuss about the general process of gaining environmental clearance of a new project.	07
8(e).	Briefly discuss about ISO 14001.	06

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